Method Preview: AdaPGC+

1 Method

1.1 Problem Setup and Overview

We consider source-free test-time adaptation for an audio–visual classifier. At test time, either modality can be corrupted or entirely missing, and no source data or target labels are available. For a test sample, the frozen backbone produces audio, video, and fused representations $\boldsymbol{z}^a, \boldsymbol{z}^v, \boldsymbol{z}^F \in \mathbb{R}^d$ whenever the corresponding views are observable. We adapt only the first fusion-attention projections and LayerNorm affine parameters, which keeps the update lightweight and limits catastrophic drift.

As illustrated in Fig. 1, AdaPGC+has three components. First, online class-conditional Gaussian models provide stable probabilistic guidance for each representation. Second, when one modality is missing, we recover the distribution of the unavailable fused representation from the observed modality. Third, we detect the less reliable modality for each sample and align only that branch, avoiding contamination of the reliable branch.

1.2 Online Probabilistic Guidance

Class-conditional modeling. For each view $r \in \{a, v, F\}$, we maintain an online Gaussian mixture with one component per class:

$$p_r(\boldsymbol{z} \mid y = c) = \mathcal{N}(\boldsymbol{z}; \boldsymbol{\mu}_c^r, \boldsymbol{\Sigma}_c^r), \qquad p_r(y = c) = \pi_c^r. \tag{1}$$

Let the pretrained linear classifier be $s_c(\boldsymbol{z}) = \boldsymbol{w}_c^\top \boldsymbol{z} + b_c$. We initialize $\boldsymbol{\mu}_c^r = \boldsymbol{w}_c$, $\boldsymbol{\Sigma}_c^r = \boldsymbol{I}$, and

$$\pi_c^r \propto \exp\big(b_c + \tfrac{1}{2}\|\boldsymbol{w}_c\|_2^2\big), \tag{2}$$

so the initial Gaussian discriminant is consistent with the source classifier. For each incoming batch, the network posterior $q_{ic} = \mathrm{softmax}(\boldsymbol{s}_i)_c$ acts as a soft assignment. We update the effective count, first moment, and second moment online:

$$\Delta N_c = \sum_i q_{ic}, \qquad \Delta \boldsymbol{S}_c = \sum_i q_{ic}\boldsymbol{z}_i, \qquad \Delta \boldsymbol{Q}_c = \sum_i q_{ic}\boldsymbol{z}_i\boldsymbol{z}_i^\top. \tag{3}$$

and use exponential smoothing to obtain π_c^r, $\boldsymbol{\mu}_c^r$, and $\boldsymbol{\Sigma}_c^r$. This soft update uses every test sample while avoiding hard pseudo-label errors.

The resulting Gaussian discriminant score is

$$\begin{aligned} g_{r,c}(\boldsymbol{z}) = {} & \log \pi_c^r - \tfrac{1}{2}\log|\boldsymbol{\Sigma}_c^r| \\ & - \tfrac{1}{2}(\boldsymbol{z} - \boldsymbol{\mu}_c^r)^\top (\boldsymbol{\Sigma}_c^r)^{-1}(\boldsymbol{z} - \boldsymbol{\mu}_c^r), \end{aligned} \tag{4}$$

where class-independent constants are omitted.

Guided adaptation. Let $\boldsymbol{g}_i$ denote the probabilistic score selected according to the sample's available modalities; its missing-modality form is defined below. We transfer this distribution to the network with

$$\begin{aligned} \mathcal{L}_g &= -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} p_{ic}^g \log p_{ic}^\theta, \\ \boldsymbol{p}_i^g &= \mathrm{softmax}(\boldsymbol{g}_i), \qquad \boldsymbol{p}_i^\theta = \mathrm{softmax}(\boldsymbol{s}_i). \end{aligned} \tag{5}$$

Only the query, key, and value projections in the first fusion-attention block are updated by this loss.

1.3 Conditional Fusion Recovery

Prediction from a single observed modality is weaker because the fused representation is unavailable. We therefore estimate the class-wise cross-covariance $\boldsymbol{\Sigma}_c^{mF} = \mathrm{Cov}(\boldsymbol{z}^m, \boldsymbol{z}^F \mid y = c)$ from fully observed test samples, where $m \in \{a, v\}$. These statistics are updated online with the same soft assignments as Eq. (3).

Given an observed feature $\boldsymbol{z}^m$ and a possible class c, Gaussian conditioning yields a recovered fused mean

$$\hat{\boldsymbol{z}}_c^{F|m} = \boldsymbol{\mu}_c^F + (\boldsymbol{\Sigma}_c^{mF})^\top (\boldsymbol{\Sigma}_c^m)^{-1}(\boldsymbol{z}^m - \boldsymbol{\mu}_c^m). \tag{6}$$

Because c is unknown, we compute a soft class responsibility from the observed view,

$$\alpha_c^m = \mathrm{softmax}(\boldsymbol{g}_m(\boldsymbol{z}^m)/T)_c\,, \tag{7}$$

score every conditional recovery with the fused-view discriminator, and marginalize over the recovery class:

$$g_{m\to F,k}(\boldsymbol{z}^m) = \sum_{c=1}^{C} \alpha_c^m \, g_{F,k}\Big(\hat{\boldsymbol{z}}_c^{F|m}\Big)\,. \tag{8}$$

Thus, the method does not commit to a single pseudo-label during recovery. The final probabilistic guidance is

$$\boldsymbol{g}_i = \begin{cases} \boldsymbol{g}_F(\boldsymbol{z}_i^F), & \text{full}, \\ (1-\beta)\boldsymbol{g}_{m\to F}(\boldsymbol{z}_i^m) + \beta \boldsymbol{g}_m(\boldsymbol{z}_i^m), & m \text{ only}. \end{cases} \tag{9}$$

Before enough paired samples have been accumulated, we safely fall back to $\boldsymbol{g}_m(\boldsymbol{z}^m)$.

1.4 Asymmetry Rectification

When both modalities are present, corruption is often asymmetric: forcing the two branches to learn from each other equally can degrade the cleaner branch. For each sample, we compare each unimodal prediction with the fused prediction using symmetric KL divergence,

$$D_m = \tfrac{1}{2}\left[\mathrm{KL}(\boldsymbol{p}_F \| \boldsymbol{p}_m) + \mathrm{KL}(\boldsymbol{p}_m \| \boldsymbol{p}_F)\right], \qquad m \in \{a, v\}. \tag{10}$$

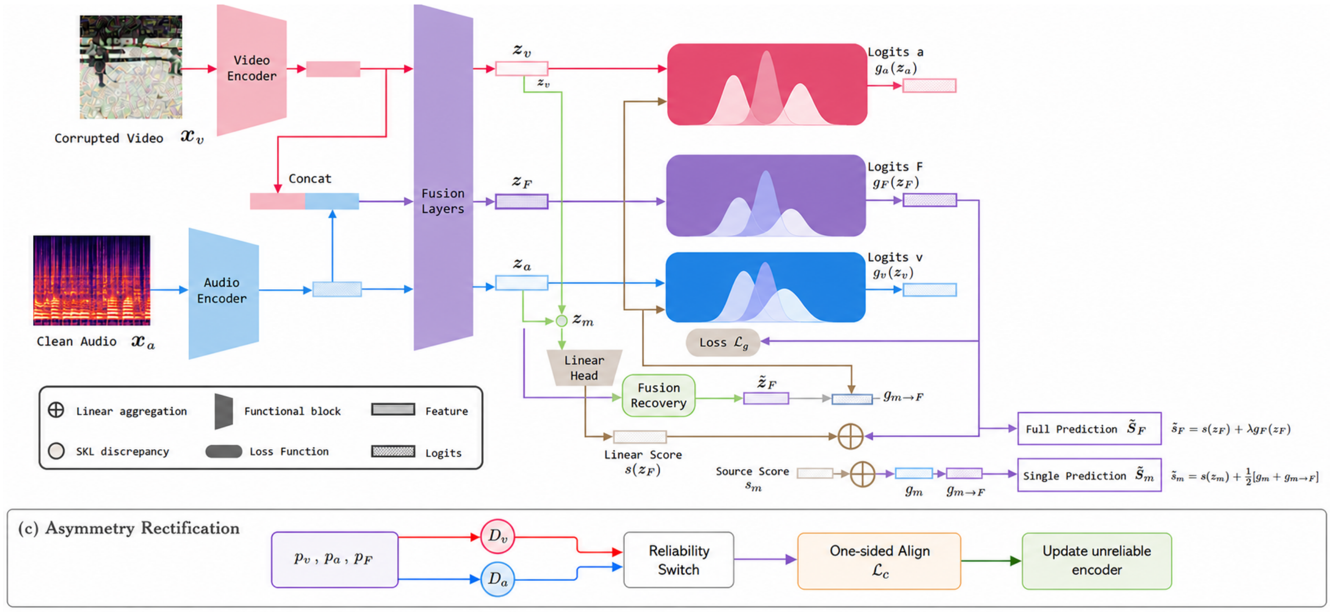

Figure 1: Overview of AdaPGC+. Online class-conditional Gaussian models guide prediction and adaptation. Fully observed samples update cross-view statistics; these statistics recover fused predictions when one modality is missing. A reliability switch performs one-sided cross-modal alignment.

The modality with the smaller D_m is treated as the reliable target, while the other modality is updated. Let r_i and u_i be the reliable and unreliable modalities for sample i. We use the one-sided contrastive loss

$$\mathcal{L}_c = -\frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \log \frac{\exp(\mathrm{sim}(\boldsymbol{z}_i^{u_i}, \mathrm{sg}(\boldsymbol{z}_i^{r_i}))/\tau)}{\sum_j \exp\big(\mathrm{sim}(\boldsymbol{z}_i^{u_i}, \mathrm{sg}(\boldsymbol{z}_j^{r_i}))/\tau\big)}, \quad (11)$$

where $\mathrm{sg}(\cdot)$ stops the gradient. Consequently, only the unreliable representation is pulled toward the reliable one. This loss updates the LayerNorm affine parameters and is applied only to fully observed samples.

1.5 Online Update and Prediction

The two parameter groups are optimized separately:

$$\mathcal{L}_{\mathrm{fusion}} = \lambda_g \mathcal{L}_g + \lambda_b \mathcal{L}_{\mathrm{base}}, \qquad \mathcal{L}_{\mathrm{LN}} = \lambda_c \mathcal{L}_c, \quad (12)$$

where $\mathcal{L}_{\mathrm{base}}$ is an optional entropy-and-diversity regularizer and can be disabled. We maintain an exponential moving average $\bar{\theta}$ of the adapted network. The final prediction combines its logits with the probabilistic guidance:

$$\widetilde{\boldsymbol{s}}_i = \boldsymbol{s}_{\bar{\theta}}(\boldsymbol{x}_i) + \gamma \boldsymbol{g}_i, \qquad \widehat{y}_i = \arg\max_c \widetilde{s}_{i,c}. \quad (13)$$

All statistics and parameters are updated once per arriving batch; no target labels, source samples, or offline retraining are required.